Pete VanBenthuysen

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Education

Johns Hopkins University — Baltimore, MD
Master of Science in Data Science — *Expected 2027*

Arizona State University — Tempe, AZ
Bachelor of Science in Data Science (Minor: Economics), magna cum laude

Summary

Experienced Data Scientist with a strong background in sports analytics, machine learning, statistical modeling, database architecture, and data quality automation. Skilled in developing and deploying predictive models, managing large-scale structured and unstructured datasets, and building scalable data pipelines that drive actionable insights and operational efficiency. Proven ability to bridge technical modeling with business and solve complex problems using innovative algorithms, while optimizing code for maximum efficiency. For a deeper look at my end-to-end project work, please visit my LinkedIn—or feel free to reach out directly.

Technical Skills

Languages: Python, R, SQL, JavaScript
Machine Learning & Deep Learning: Scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, neuralnet
Statistical Tools & Methods: Time series analysis, multivariate regression, Monte Carlo simulation, Markov chains
Data Visualization: Plotly, Seaborn, Matplotlib, ggbiplot, Tableau, PowerBI, Alteryx
Databases: PostgreSQL, IBM DB2
Developer Tools & Platforms: Git, Docker, Kubernetes, AWS (S3, EC2, Lambda), VSCode, PyCharm, Collibra DQ
AI / LLMs: ChatGPT/OpenAI API wrappers for summarization, note-taking, and workflow automation
Licenses and Certifications: AWS Cloud Practitioner Candidate, Microsoft Office Specialist (Word, Excel, PowerPoint)
Leadership and Community Involvement: NLHS Society, Dean's list, ASU Devil Data Science, ASU CodeDevils

Research

Clutch Geometry: Survival Analysis of 1vX in CS2 | *Python, survival/hazard modeling, spatial features* — 2025 – Present

- Built a CS2 data scraper and telemetry pipeline (modular scrapers + parsers) to collect round/second-level positions, utility, and map-zone geometry used for modeling
- Developed a second-by-second survival model for 1vX clutches using spatial geometry (angles, isolations, lines of sight, post-plant utility)
- Selected to present at the Sloan Sports Analytics Conference (March 2026) to an audience of team and league executives

Projects

Predictive Data Quality Forecasting System | *XGBoost, Time Series, PostgreSQL, Power BI* — Jun. 2025 – Jul. 2025

- Developed a scalable pipeline to forecast three months of KDE-level data quality metrics achieving average error rates under 2% per metric
- Automated ingestion and retraining workflows using a sliding window approach, uploading results to a PostgreSQL server for internal tracking and long-term storage
- Delivered forecasted outputs to stakeholders via Power BI dashboards, enabling proactive issue detection and reducing manual remediation cycles, improving operational efficiency and cost savings

Automated NBA Data Pipeline | *Python, Playwright, SQL, AWS, API Integration* — Jan. 2025 – Present

- Built a scalable data pipeline to ingest, clean, and store NBA player and team logs, supporting over 1,000 automated daily queries across multiple seasons
- Implemented relational schema design and optimized batch inserts into SQL databases for long-term analytics
- Deployed data pipeline to AWS for cloud-based access, storage, and automated daily sync with public APIs and modular scrapers

Experience

MUFG Bank — Data Quality Solutions — Tempe, AZ
Data Scientist — Financial Crimes Division — *May 2025 – Present*

- Design and deploy a forecasting pipeline for rule-level data-quality breakage, enabling proactive decisions across the enterprise
- Partner with business teams to triage JIRA issues and diagnose upstream/downstream breaks across complex pipelines, reducing resolution times
- Author SQL to extract and normalize malformed XML within enterprise data stores; align outputs to governance and compliance standards